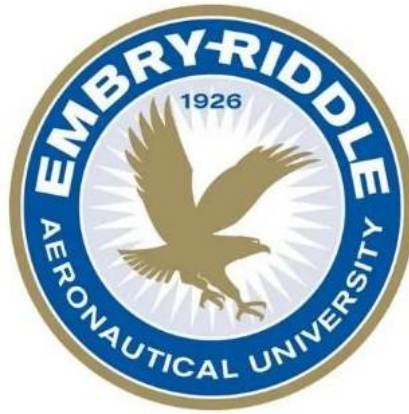




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AE 399: Special Topic in Aerospace Engineering

Course Topic: Turbomachinery Cascade Rig

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April 25, 2026

## Introduction:

Accurate prediction and measurement of flow behavior in turbine blade cascades is critical for aerodynamic performance analysis and design validation. In the current Gas Turbine Laboratory setup, flow measurements are obtained using a single-point pitot probe, which limits spatial resolution and requires multiple test runs to map the flow field.

This project addresses both the computational and experimental limitations of the existing system. The work began with the development of an Excel-based analytical framework to compute turbine blade deviation angles and throat openings, followed by automation using Python. In parallel, a multi-point probe rake system was designed and fabricated to improve experimental measurement capability.

## Objectives:

- Develop a structured method to compute deviation angles ( $\delta$ ) and throat openings ( $\phi$ )
- Implement Aungier's empirical turbine correlations
- Automate the calculation process using Python
- Improve experimental measurement capability with a multi-point probe system
- Design and prototype a custom probe rake for the cascade rig

## Computational Development:

The initial phase of the project focused on building a structured computational model in Excel to determine turbine blade parameters following Aungier's methodology, i.e., sections 4.1 and 4.2 of Chapter 4 "Empirical Performance Models For Axial-Flow Turbine Blade Rows" from Aungier's Turbine Aerodynamics.

### I. Initial Assumption Table

An initial table was constructed using the assumptions:

- Deviation angle:  $\delta = 0$  for all blade rows
- Mach number:  $M = 1$  at blade exit for all blade rows

This allowed computation of baseline values assuming ideal flow conditions. For each blade row (S1, R1, S2, R2):

Pitch:

$$s = \frac{2\pi r}{N} \quad (1)$$

Gauging angle:

$$\beta_g = \alpha_2 \quad (2)$$

Throat opening:

$$o = s \cdot \sin(\beta_g) \quad (3)$$

This step established a reference dataset for further refinement.

## II. Variable Deviation Angle Table

A second table was created where  $\delta$  was treated as a variable input.

Gauging angle updated as:

$$\beta_g = \alpha_2 - \delta \quad (4)$$

Throat opening recomputed:

$$o = s \cdot \sin(\beta_g) \quad (5)$$

| deviation angle = 0 & Mach = 1 |              |    |             |                        |                    |
|--------------------------------|--------------|----|-------------|------------------------|--------------------|
| Blade                          | Radius (mid) | N  | Pitch (s)   | Gauging angle (Beta g) | Throat opening (o) |
| S1                             | 0.254647909  | 57 | 0.028070175 | 56.85511969            | 0.023502897        |
| R1                             | 0.254647909  | 47 | 0.034042553 | 51.75291049            | 0.026735264        |
| S2                             | 0.233625606  | 53 | 0.027696471 | 61.79303862            | 0.024407405        |
| R2                             | 0.223574707  | 49 | 0.028668598 | 51.54393218            | 0.022449956        |

Figure 1: Excel example with  $\delta = 0^\circ$

| deviation angle | 4.580836048  |    |             |                        |                    |
|-----------------|--------------|----|-------------|------------------------|--------------------|
| Blade           | Radius (mid) | N  | Pitch (s)   | Gauging angle (Beta g) | Throat opening (o) |
| S1              | 0.254647909  | 57 | 0.028070175 | 52.27428364            | 0.022202077        |
| R1              | 0.254647909  | 47 | 0.034042553 | 47.17207444            | 0.024966762        |
| S2              | 0.233625606  | 53 | 0.027696471 | 57.21220257            | 0.023283924        |
| R2              | 0.223574707  | 49 | 0.028668598 | 46.96309613            | 0.020954288        |

Figure 2: Excel example with variable  $\delta$

This allowed observation of how the deviation angle affects geometry and flow.

## III. Deviation Angle Using Aungier Correlations

Deviation angles were then computed using appropriate equations from Aungier based on the exit Mach number:

- $M = 1 \rightarrow \text{Eq. 4.14}$

$$\delta = \arcsin\left(\frac{o}{s} \times \frac{\rho_* W_*}{\rho_2 W_2}\right) - \beta_g \quad (6)$$

- $M > 0.5 \rightarrow \text{Eq. 4.12, 4.13, 4.10}$

$$\delta = \delta_0 [1 - 10X^3 + 15X^4 - 6X^5] \quad (7)$$

$$X = 2M_{exit} - 1 \quad (8)$$

$$\delta_0 = \arcsin\left(\frac{o}{s} \left[1 + \left(1 - \frac{o}{s}\right) \left(\frac{\beta_g}{90}\right)^2\right]\right) - \beta_g \quad (9)$$

- $M \leq 0.5 \rightarrow \text{Eq. 4.11}$

$$\delta = \delta_0 \quad (10)$$

Each blade row required a different equation depending on its flow condition.

#### IV. Goal Seek Implementation

To determine the correct deviation angle:

- A residual was defined:

$$\text{Residual} = \delta_{\text{guess}} - \delta_{\text{calculated}} \quad (11)$$

- Excel Goal Seek was used:

- Set residual  $\rightarrow 0$
- Adjust  $\delta$

This produced the final converged deviation angle for each blade.

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Figure 3: Excel example with variable  $\delta$  for the Goal Seek application

#### V. Final Throat Opening Calculation

Using converged  $\delta$  values:

- Final gauging angle:

$$\beta_g = \alpha_2 - \delta \quad (12)$$

- Final throat opening:

$$o = s \cdot \sin(\beta_g) \quad (13)$$

|  |    | Pitch (s)   | Deviation angle (del) | Gauging angle (Beta g) | Throat opening (o) |
|--|----|-------------|-----------------------|------------------------|--------------------|
|  | S1 | 0.028070175 | 0                     | 56.85511969            | 0.023502897        |
|  | R1 | 0.034042553 | 4.580844766           | 47.17206573            | 0.024966758        |
|  | S2 | 0.027696471 | 2.078804171           | 59.71423445            | 0.023916481        |
|  | R2 | 0.028668598 | 4.689561658           | 46.85437052            | 0.020917122        |

Figure 4: Final Calculation Table Example

This completed the full computational process

## Python Automation:

After validating the Excel workflow, the process was automated using Python. The key features include:

- Reads all input data directly from Excel
- Reproduces calculations automatically
- Uses numerical root-finding (fsolve) instead of Goal Seek
- Updates outputs instantly when Excel inputs change

```

11 # Loading excel file
12 df = pd.read_excel('TurboDesigns.xlsx', sheet_name='T20250508', header=None)
13
14 #####
15
16 # Constants
17
18 # Mid Radius

```

```

(base) PS C:\Users\anura\OneDrive - Embry-Riddle Aeronautical University\Desktop\GAS TURBINE LAB> & c:\Users\anura\AppData\Local\Programs\Python\Python313\python.exe "c:\Users\anura\OneDrive - Embry-Riddle Aeronautical University\Desktop\GAS TURBINE LAB\turbineCode.py"
Throat Opening (o) for S1 : 0.0235 meters
Throat Opening (o) for R1 : 0.0250 meters
Throat Opening (o) for S2 : 0.0239 meters
Throat Opening (o) for R2 : 0.0209 meters
(base) PS C:\Users\anura\OneDrive - Embry-Riddle Aeronautical University\Desktop\GAS TURBINE LAB>

```

Figure 5: Python Output Example

The script imports data from the Excel file and can reproduce new values automatically whenever the Excel inputs are updated. The program is also modular, allowing users to change parameters easily and accessibly.

## Experimental System Limitation

The existing cascade rig uses a single pitot probe, which functions:

- One-point measurement at a time
- Multiple runs required for full mapping
- Low spatial resolution
- Inefficient testing process

This created the need for a multi-point measurement system.

## Probe Rake Design and Development

### I. Design Requirements

To enable a multi-point measurement system, the initial approach was to identify an existing pitot rake design that could be directly implemented in a laboratory setting. This involved researching commercial solutions as well as studying the underlying design principles from academic literature and manufacturer specifications. A reference design from NASA served as a baseline for understanding the structure and functionality of multi-probe rakes.

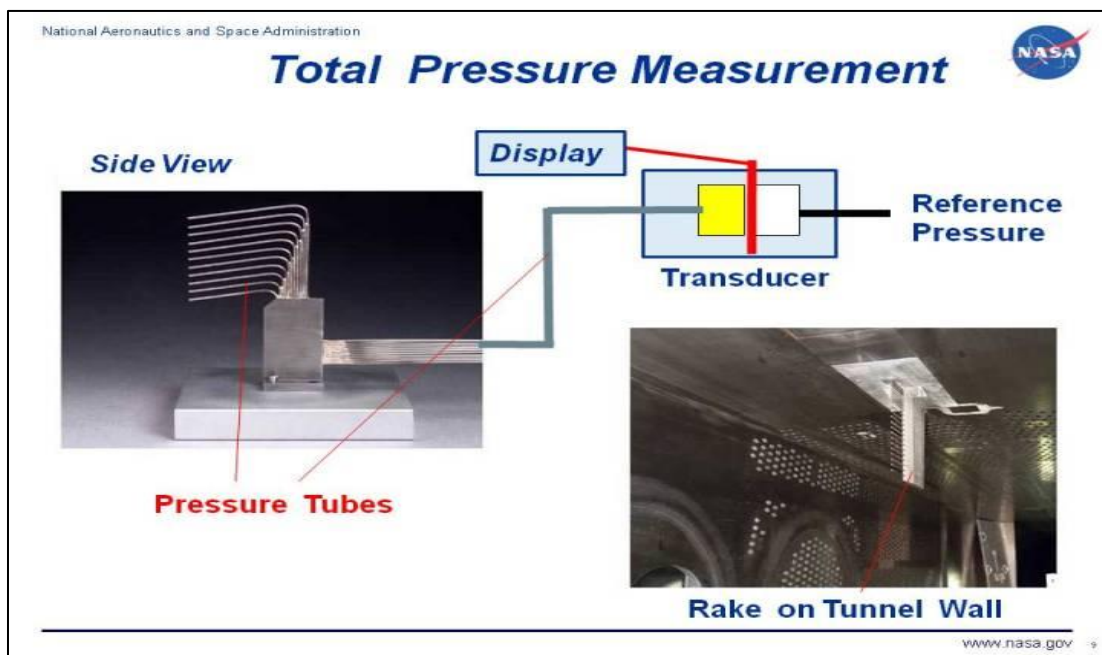
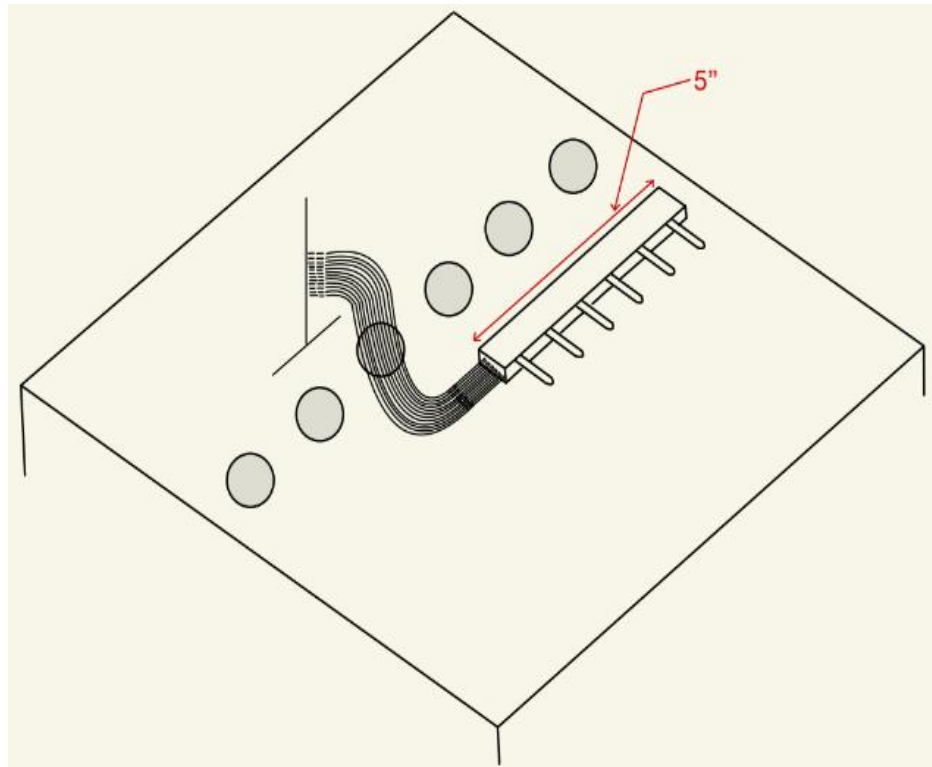


Figure 6: NASA's Pitot Tube Rake

In addition, relevant literature was reviewed, particularly “Design and Performance of a Low-Cost Pitot Rake” by Alexander Baron von Hohenhau, which provided insight into practical design considerations such as probe spacing, structural support, and measurement accuracy.

## II. Design Concept

Based on this research, an initial conceptual model was developed to fit the constraints of the existing turbine cascade apparatus. This model aimed to replicate a traditional rake configuration, where multiple pitot tubes are mounted on a single structure and inserted through the port.



*Figure 7: Draft design of the Rake*

Following this, several manufacturers were contacted to explore commercially available or custom-built solutions. However, none of the available options satisfied the specific geometric and operational requirements of the laboratory setup. This limitation led to the decision to develop a custom design using CAD.

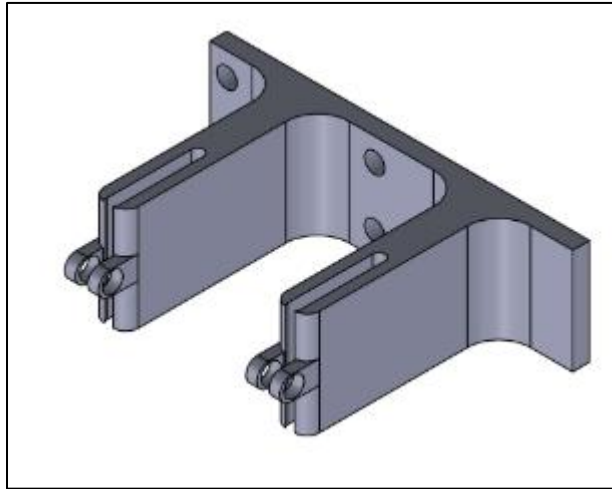
During the design process, an improved concept was identified. Instead of inserting a fully assembled rake through the port, a multi-arm support structure was designed to be mounted externally above the glass top of the cascade. This structure supports multiple pitot tubes while allowing independent vertical adjustment through the existing mechanism. This approach offers significant advantages, including simplified installation, improved accessibility, and easier maintenance, as it eliminates the need to remove the top section of the rig.

The detailed CAD drawings (see Appendix) define all geometric parameters; however, the key design requirements included:

- A support body length of approximately 18 inches
- A configuration of five arms to accommodate five pitot tubes
- Alignment with the existing port hole spacing

### III. CAD Development

For the initial iteration, a two-arm prototype was fabricated to evaluate dimensional accuracy and material strength. This prototype was used to assess the structural capability of the design and verify fit within the system.



*Figure 8: CAD model – 1<sup>st</sup> Iteration*

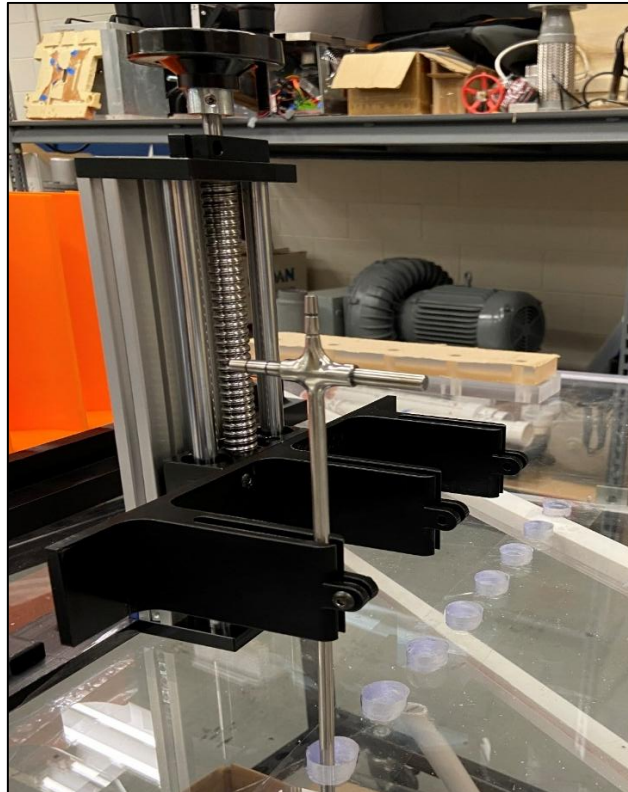
### IV. Fabrication - 3D Printing

The model was prototyped using 3D printing, using PETG as the material:

- Enabled rapid iteration
- Allowed testing of fit and geometry
- Adjusted tolerances for tube insertion



*Figure 9: Lab Apparatus – 2 Support Arms*



*Figure 10: Lab Apparatus – 3 Support Arms*

## Final Outcome

Following iterative refinement of arm spacing, tolerances, and structural stability, a final five-arm support structure was fabricated using 3D printing. The completed design successfully supports multiple pitot tubes and enables simultaneous multi-point measurements, significantly improving the experimental capability of the turbine cascade rig. The project resulted in:

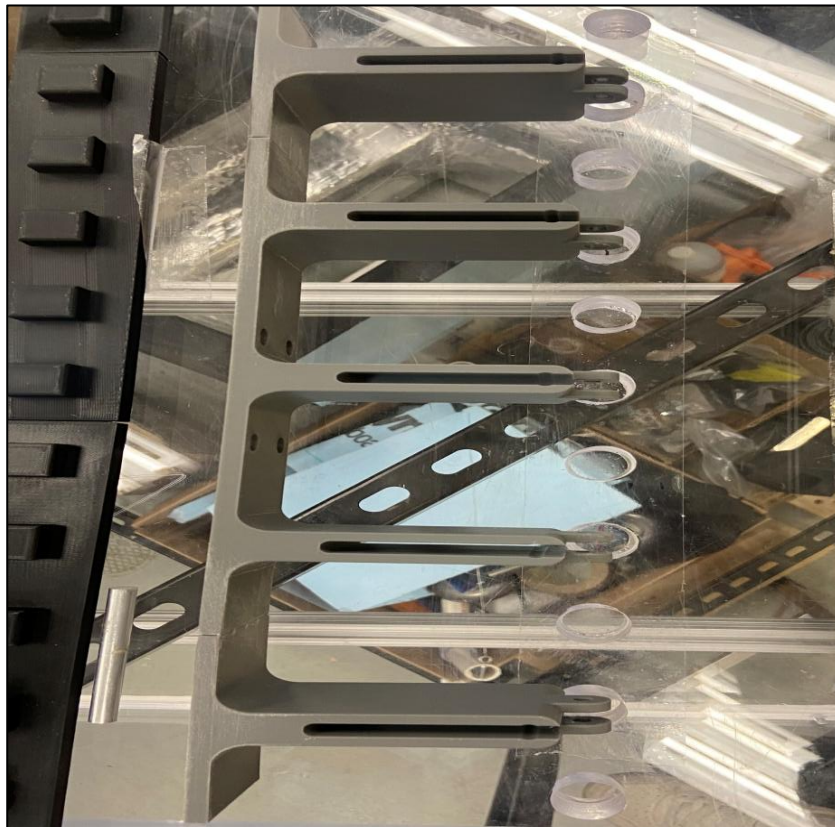
- A validated Excel-based computational framework
- A fully automated Python analysis tool
- A functional multi-point probe rake design



*Figure 11: Current Lab Apparatus – 3 Support Arms*



*Figure 12: Proposed Lab Apparatus – 5 Support Arms*



*Figure 13: Proposed Lab Apparatus – 5 Support Arms*

This project successfully developed an integrated workflow combining Excel-based modeling, Python automation, and experimental system design. The work improves both the accuracy of turbine parameter prediction and the efficiency of experimental data collection, providing a strong foundation for future turbine aerodynamics research.

## **Appendix**



